MILITARY SPECIFICATION

BOLT ASSEMBLY, CLAMP, AND COMPONENT
PARTS, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers clamp bolt assemblies and component parts used for weapon system containers.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-P-416 - Plating, Cadmium (Electrodeposited)
PPP-H-1581 - Hardware (Fasteners and Related Items), Packaging of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Armament Research and Development Center, U S Army Armament, Munitions and Chemical Command, Attn. DRSMC-TST-S (D), Dover, NJ 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-B-70332/1	-	Bolt Assembly, Clamp
MIL-B-70332/2	-	Bolt, Shoulder, (Bolt Assembly, Clamp)
MIL-B-70332/3	-	Bracket, Stop (Bolt Assembly, Clamp)
MIL-B-70332/4	-	Washer, Shouldered (Bolt Assembly, Clamp)
MIL-B-70332/5	-	Washer, Spring Tension (Bolt Assembly, Clamp)

STANDARDS

FEDERAL

FED-STD-H28/2	-	Screw Thread Standards for Federal Services, Section 2, Unified Thread Form and Thread Series for Bolts, Screws, Nuts, Tapped Holes and General Applications
FED-STD-H28/20	-	Inspection Methods for Acceptability of UN, UNR, UNJ, M and MJ Screw-Threads

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MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	-	Identification Marking of U. S. Military Property
MIL-STD-1312	-	Fasteners, Test Methods
MS24665	-	Pin, Cotter (Split)

AIR FORCE - NAVY AERONAUTICAL

AN121554	-	Nut - Castle

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be of the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A29	-	Steel Bars, Carbon and Alloy, Hot-Wrought and Cold-Finished, General Requirements for
ANSI/ASTM A505	-	Steel Sheet and Strip, Alloy, Hot-Rolled and Cold Rolled, General Requirements for
ASTM A510	-	General Requirements for Wire Rods and Coarse Round Wire, Carbon Steel
ANSI/ASTM A659	-	Steel, Carbon (0.16 Maximum to 0.25 Maximum, Percent), Hot-Rolled Sheet and Strip, Commercial Quality

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(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

AMS 6350 - Steel Sheet, Strip, and Plate 0.95Cr - 0.20Mo (0.30 - 0.35C)
AMS 6355 - Steel Sheet, Strip, and Plate 0.50Cr - 0.55Ni- 0.20Mo (0.28 - 0.33C) (SAE 8630)
AMS 6356 - Steel Sheet, Strip, and Plate 0.95Cr - 0.20Mo (0.28 -0.33C) (SAE 4130)

(Applications for copies should be addressed to The Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references, other than specification sheets cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 Material. Recycled and reclaimed materials shall be used to the maximum extent practicable. Components shall be made of alloy steel or carbon steel of a grade specified in the individual specification sheet. specification sheet.

3.3 Design.

3.3.1 Threads. Component threads shall conform to the unified thread form and class after plating, as specified on the applicable specification sheet and as specified in FED-STD-H28/2 (acceptability of screw threads shall be in accordance with FED-STD-H28/20, System 21).

3.4 Mechanical properties. Mechanical properties for each component shall be as specified on the applicable specification sheet.

3.4.1 Hardness. Components shall be within the Rockwell hardness range when specified on the applicable specification sheet and tested in accordance with 4.5.4.

3.4.2 Tensile strength. Bolts shall have the tensile strength specified on the applicable specification sheet when tested in accordance with 4.5.2.

3.5 Protective finish. Components shall be cadmium plated in accordance with QQ-P-416, Type II, Class 2 as specified in the applicable specification sheet.

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3.6 Identification marking. Components shall be marked in accordance with MIL-STD-130.

3.7 Workmanship. Finished components shall be uniform in quality and appearance. Surfaces shall be free of foreign matter. Any surfaces in the unplated condition shall be free of cracks, bursts, folds, tool marks, seams, voids on the bearing surface, nicks and gouges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Quality conformance inspection shall be as specified in Table I.

TABLE I. QUALITY CONFORMANCE INSPECTION.

EXAMINATION OR TEST	REQUIREMENT PARAGRAPH	EXAMINATION OR TEST PARAGRAPH
Group A Examination	1/	4.5.1
Group B Mechanical properties Protective finish	 3.4 3.5	 4.5.2 4.5.3

1/ All requirements not covered by tests.

4.3 Materials inspection. Materials inspection shall consist of certification supported by verifying data that the materials used in fabricating the components are in accordance with 3.2, 3.4 and 3.5.

4.4 Inspection lot. An inspection lot shall consist of all items covered by a single specification sheet, produced under essentially the same conditions, and offered for inspection at one time.

4.4.1 Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

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4.4.2 Sampling for Group A inspection. Sampling shall be in accordance with MIL-STD-105, Inspection Level II, with an Acceptable Quality Level (AQL) of 1.5 percent defective for major defects. For minor defects, Inspection Level I with an AQL of 4.0 percent defective shall apply.

4.4.2.1 Classification of defects. Defects shall be classified as specified in Table II.

TABLE II. CLASSIFICATION OF DEFECTS.

CATEGORY	DEFECT	INSPECTION METHOD
Critical	None defined	
Major		
101	Bolt threads not as specified (see 3.3.1)	Gage
102	Width across flats not as specified (see 3.1)	SIE 1/
103	Bolt shoulder diameter not as specified (see 3.1)	SIE
104	Bracket and washer inside hole diameters not as specified (see 3.1)	SIE
105	Evidence of cracks, flaws or pits	Visual
106	Bracket and washer thickness not as specified (see 3.1)	SIE
107	Tensile strength not as specified (see 4.5.2)	SIE
108	Hardness not as specified (see 4.5.4)	SIE
Minor		
201	Overall length of bolt not as specified (see 3.1)	SIE
202	Bolt shoulder length not as specified (see 3.1)	SIE
203	Other dimensions not as specified (see 3.1)	SIE
204	Protective finish missing or incomplete, when required (see 3.5)	Visual
205	Workmanship not as specified (see 3.7)	Visual
206	Packaging not as specified (see 5.1)	Visual

1/ Standard inspection equipment.

4.4.3 Sampling for Group B inspection.

4.4.3.1 Sampling for mechanical properties tests. Sampling shall be in accordance with MIL-STD-105 Inspection Level S-2 with an AQL of 1.5 percent defective.

4.4.3.2 Sampling for protective finish test. Sampling for protective finish test shall be in accordance with applicable specifications referenced in 3.5.

4.4.3.3 Disposition of sample units. Items which have been subjected to Group B inspection shall not be delivered on the contract.

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4.4.4 Packaging inspection. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of PPP-H-1581.

4.5 Methods of inspection.

4.5.1 Examination. Sample items shall be thoroughly examined to determine conformance with this specification and applicable specifications with respect to all the requirements not covered by tests.

4.5.2 Bolt mechanical properties. Sample bolts shall be tested for conformance to requirements of 3.4. Tensile strength shall be determined in accordance with MIL-STD-1312, Test Method 8.

4.5.3 Protective finish. Samples taken as specified in 4.4.3.2 shall be inspected for adequacy of plating in accordance with the applicable specification of 3.5.

4.5.4 Surface hardness. Samples taken as specified in 4.4.3.1 shall be tested for surface hardness in accordance with MIL-STD-1312, Test Method 6.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with PPP-H-1581 (see 6.2).

6. NOTES

6.1 Intended use. Clamp bolt assemblies are intended for use as part of weapon system containers.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification and the applicable specification sheet.
b. Applicable specification sheet part number (see 3.1).
c. Level (degree) of protection, in accordance with PP-H-1581, ordering data (see 5.1).

Custodians:
Army - AR
Navy - AS
Air Force - 99

Preparing activity:
Army - AR

(Project 53GP-0131)

Review activities:
Army - MI
Navy - OS
Air Force - 82

Agent:
DLA - IS

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